

Model-free control methodology and applications: from aerodynamics to neurosciences

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Seminar KTH - Tuesday 12 August

Outline

- Overview of the model-free control methodology
- Applications:
 - ▶ Aerodynamic lift tracking in wind turbines
 - ▶ HIV epidemic propagation
 - ▶ Pneumatic-muscle-based gripper task operation
 - ▶ Growing neural network closed-loop learning

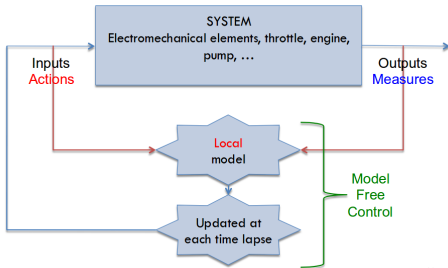
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Overview of Model-Free Control

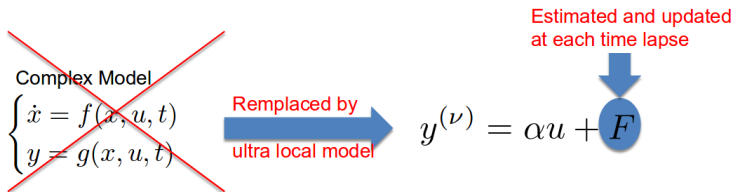
(Fliess & Join - 2008)

- From a complex plant... to a simple local approximation: a *local model* is derived at each instant



Model-Free Control is based on the control of the local model (that compensates the poorly known parts of the plant as well as of the various possible disturbances)

Principle of Model-Free Control



Ultra local model control based

$$u = \frac{1}{\alpha} \left(-F + y_*^{(\nu)} \right) + \text{Correction}(e)$$

e : tracking error

(c) Pr Cédric Join

- Same conceptual simplicity as PID controllers
- No need for precise modeling / identification
- Requires the evaluation of time derivatives of noisy signals

Model-free-based control algorithms

- **(Fliess, Join - 2008)**¹ The *model-free control* algorithm can be considered as an extended PID controller that uses a very simple local approximation of any complex system
- **(Michel - 2012)**² A derivative-free version has been proposed as a "self-tuned integrator"

1. *Int. J of Control*, 2013 & *Int. J of Robust and Nonlinear Control*, 2021
presentation available at <https://mps2016.labri.fr/archives/join.pdf>

2. <https://arxiv.org/abs/1202.4707>

A derivative-free version: Para-Model Control

(Michel - 2012)

Para-Model Control (PMC)

The discrete controller $u = PMC(y, y^*)$ reads:

$$u_k = \Psi_k \cdot \sum_{j=1}^k K_i(y_j^* - y_j)h, \quad h > 0$$

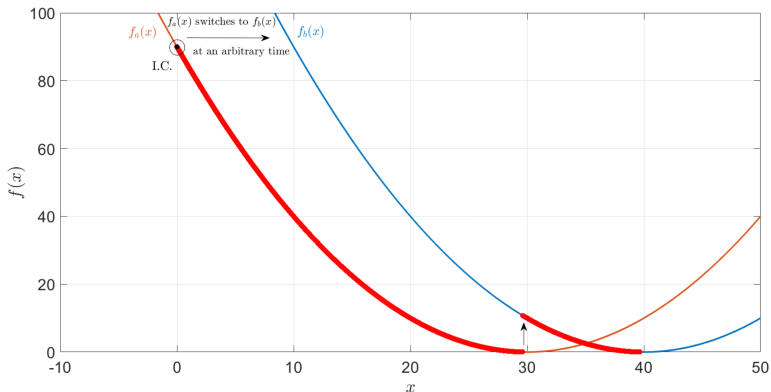
with $\Psi_k = \Psi_{k-1} + K_p(k_\alpha e^{-k_\beta k} - y_k)$

where : y^* is the output reference trajectory ; K_p , K_i , k_α and k_β are tuning gains and $k > 0$ denotes the discrete time index

- Ψ is a time series that "adjusts" online the gain of the integrator, associated to a forgetting factor

Para-Model Control as an online optimizer

Basic example of online minimum-seeking of "unknown" switching convex functions $f_a \rightarrow f_b$ at an arbitrary time



- Stabilization *a priori* of the tracking first at the f_a then f_b min.

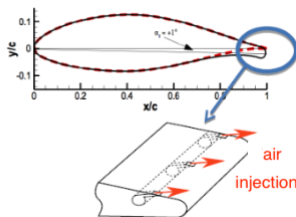
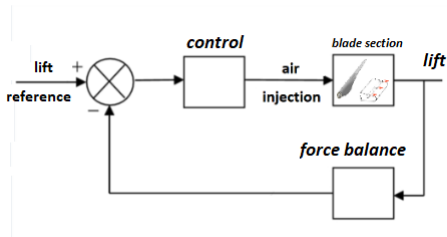
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Aerodynamic lift tracking in wind turbines

Lift tracking at the blade scale with model-free control - wind tunnel experiments

- Lift control is achieved by a control loop that drives the *pressurized air* above the blade according to the measured lift, to track the lift reference



L. Michel et al., Comparison of different feedback controllers on an airfoil benchmark, Wind Energ. Sci., 10, 177–191, 2025.

Aerodynamic lift tracking in wind turbines

Wind tunnel setup - the test section

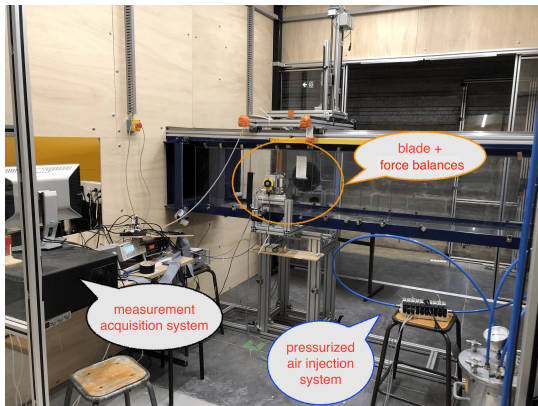
A small scale wind turbine blade is placed inside the wind tunnel



Aerodynamic lift tracking in wind turbines

Wind tunnel setup - the test section

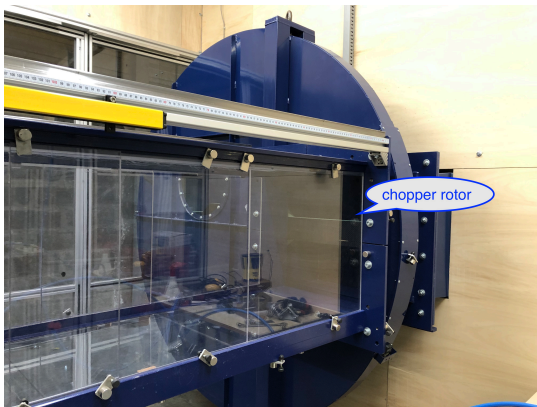
A small scale wind turbine blade is placed inside the wind tunnel



Aerodynamic lift tracking in wind turbines

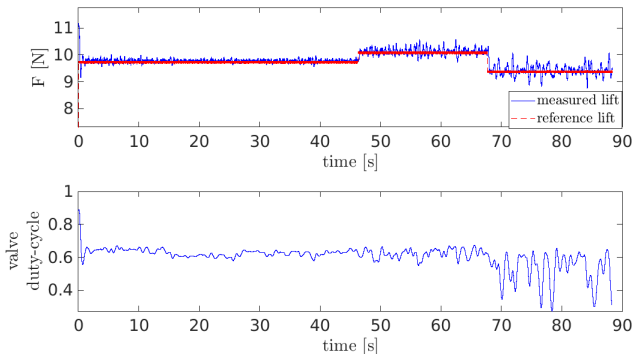
Wind tunnel setup - the chopper system

The chopper is used to disturb the airflow: strong decrease of the lift on the blade



Aerodynamic lift tracking in wind turbines

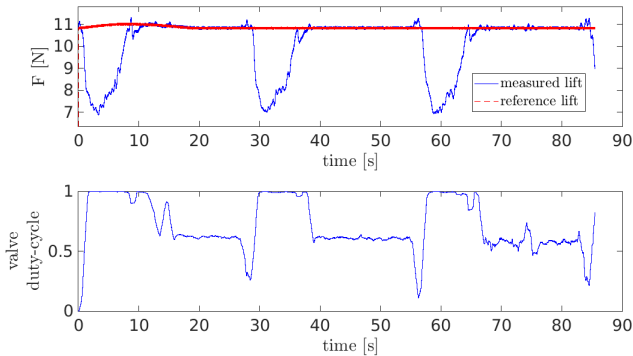
Experimental validation



- Perturbation "static chopper" from the beginning of the control with a varying reference

Aerodynamic lift tracking in wind turbines

Experimental validation

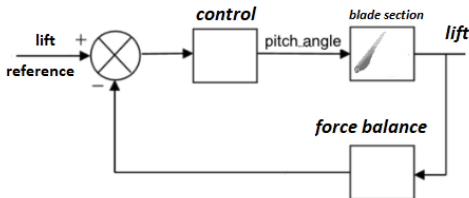


- Perturbation "rotating chopper" from the beginning of the control $\Rightarrow$ *Learning effect of the control*

Aerodynamic lift tracking in wind turbines

Lift tracking at the blade scale with para-model control - CFD validation

- Lift control is achieved by a control loop that drives the *pitch angle* according to the measured lift, to track the lift reference

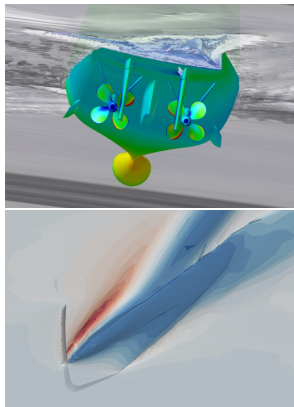


L. Michel et al., Model-free based pitch control of a wind turbine blade section: aerodynamic simulation, ECC'24.

Aerodynamic lift tracking in wind turbines

The ISIS-CFD flow solver

- Developed at Centrale Nantes/CNRS,
- Part of FINE/Marine from Cadence,
- Two-fluid Navier-Stokes,
- Face-based Finite-Volume discretisation,
- Several RANS and hybrid RANS/LES turbulence models,
- Body motion,
- Adaptive grid refinement.

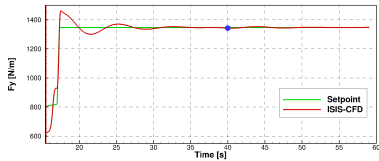
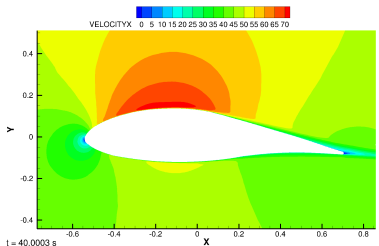
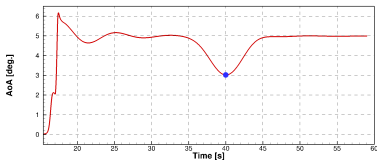
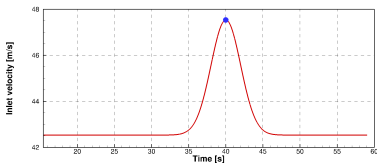


Aerodynamic lift tracking in wind turbines

Simulation results

$$F_y = 1347 \text{ n/m}, U_\infty = 42.54 \text{ m/s} (Re = 3.52 \times 10^6)$$

$t = 40.00 \text{ sec} : U_\infty = 47.54 \text{ m/s}$
 $\Rightarrow \text{AoA} = 3.02^\circ$



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HIV epidemic propagation

HIV virus minimization: preliminary results

(Craig, Xia, Venter - 2004)

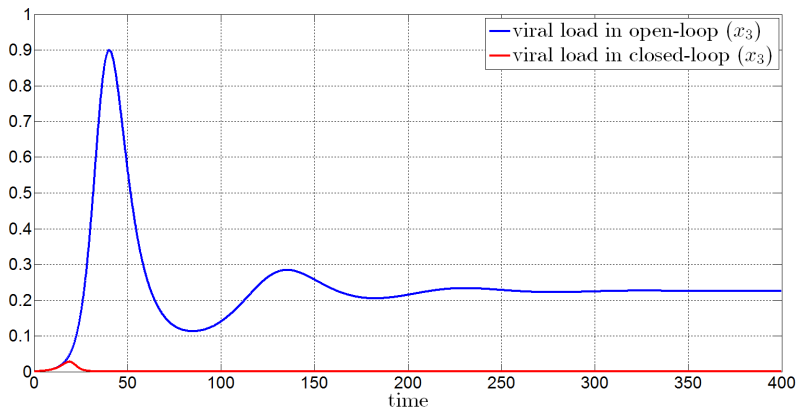
- Control of the predator-prey like model that describes the evolution of the HIV-1 dynamics when subjected to an external "medical agent"

$$\begin{cases} \dot{x}_1 = s - dx_1 - (1 - u_1)\beta x_1 x_3 \\ \dot{x}_2 = (1 - u_1)\beta x_1 x_3 - \mu x_2 \\ \dot{x}_3 = (1 - u_2)kx_2 - cx_3 \\ y = \begin{pmatrix} 0 & 0 & \gamma \end{pmatrix} x \end{cases}$$

HIV epidemic propagation

HIV virus minimization: preliminary results

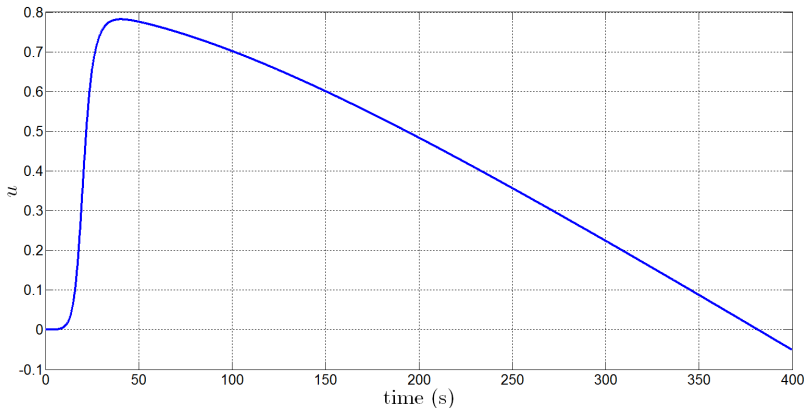
- Open-loop vs closed-loop



HIV epidemic propagation

HIV virus minimization: preliminary results

- Evolution of the associated u variable "medical agent"



HIV epidemic propagation

HIV-1 epidemiological model: towards minimizing infected cases

(Silva, Torres - 2018)

Consider the SICAE mathematical model for HIV/AIDS transmission

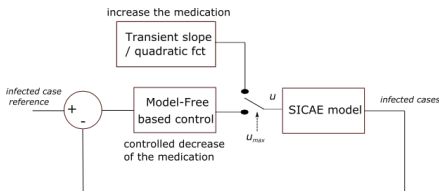
$$\left\{ \begin{array}{l} \dot{S}(t) = \mu N - \frac{\beta}{N}(I(t) + \eta_C C(t) + \eta_A A(t))S(t) - \mu S(t) - S(t)u(t) + \theta E(t), \\ \dot{I}(t) = \frac{\beta}{N}(I(t) + \eta_C C(t) + \eta_A A(t))S(t) - (\rho + \phi + \mu)I(t) + \alpha A(t) + \omega C(t), \\ \dot{C}(t) = \phi I(t) - (\omega + \mu)C(t), \\ \dot{A}(t) = \rho I(t) - (\alpha + \mu)A(t), \\ \dot{E}(t) = S(t)u(t) - (\mu + \theta)E(t). \end{array} \right.$$

where u is the *driven* medication input and I is the *controlled* infected state

HIV epidemic propagation

HIV-1 epidemiological model: control architecture

- Control sequence to drive in real-time the medication u in order to minimize the infected cases

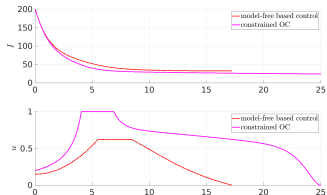


- The goal is to minimize the infected state I considering the SICAE model as a black box under the $S.u \leq 2000$ constraint

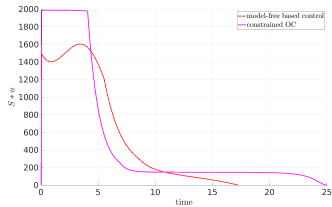
L. Michel et al., Model-free based control of a HIV/AIDS prevention model, Mathematical Biosciences Engineering, vol. 19, Issue 1, pp. 759-774, 2022.

HIV epidemic propagation

Results - controlled SICAÉ model using a quadratic increasing transient (I)



(a) Evolution of the infected state I versus the Time (year).

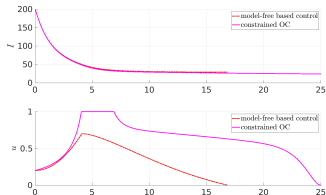


(b) Evolution of the controlled medication u versus the Time (year).

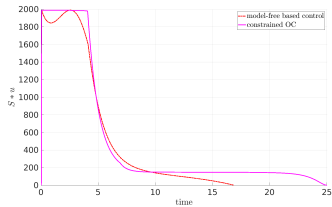
Evaluation of the **constrained** model-free SICAÉ control

HIV epidemic propagation

Results - controlled SICA model using a quadratic increasing transient (II)



(c) Evolution of the infected state I versus the Time (year).



(d) Evolution of the controlled medication u versus the Time (year).

Evaluation of the **constrained** model-free SICA model control

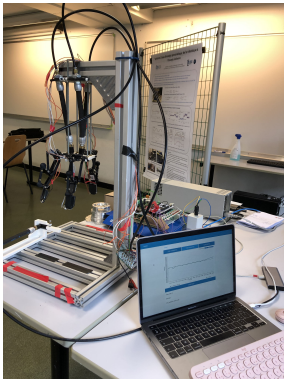
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Pneumatic-muscle-based gripper task operation

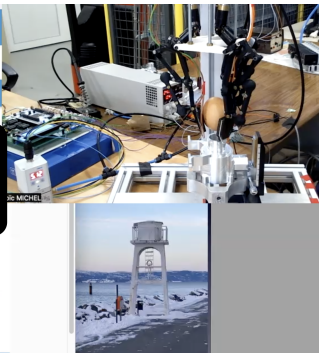
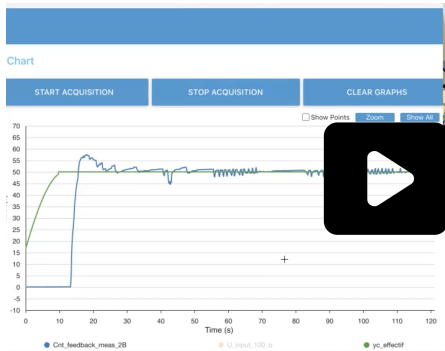
Control of the grasping effort

- No prior modeling of the kinematic motions (a pneumatic muscle is highly non-linear)
- Hybrid strategy to avoid piece detection / control of the grasping effort



Pneumatic-muscle-based gripper task operation

Experimental results



Video showing the system grasping a fragile object (e.g., an egg)...

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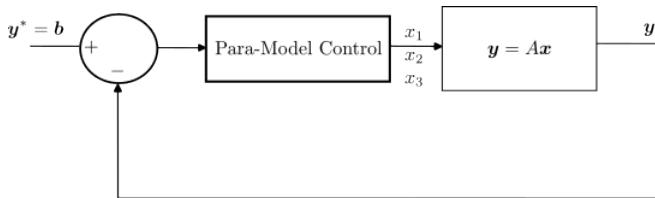
- Training a basic network
 - An amazing way to solve $Ax = b$
 - Application to a basic neural network
- Towards growing neural network: a connectome perspective
- Open perspectives

An amazing way to solve $Ax = b$ in closed-loop

Consider the following linear system $Ax = b$ to solve

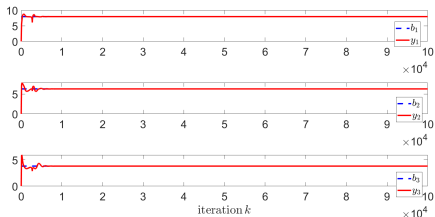
$$\begin{pmatrix} 3 & 0.5 & 8 \\ 4 & 7 & 4.5 \\ 1 & 9 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7.95 \\ 6.30 \\ 3.80 \end{pmatrix} \quad (1)$$

in closed-loop considering the vector x as the driven variable

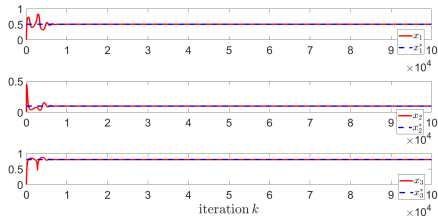


- The control $x = PMC(y, b)$ tracks $y^* = b$ and solves (1)

An amazing way to solve $Ax = b$ in closed-loop



Evolution of y vs. iterations



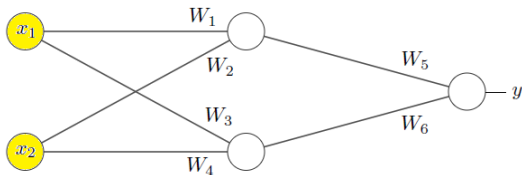
Evolution of x vs. iterations

- If y is "close" to b , then the driven x is "close" to the true solution x^*

Application to a basic neural network training

Let us apply the previous strategy to train a 3-neuron network³

Given a data training set $\{x_1^{train}, x_2^{train}, y^{train}\}$,

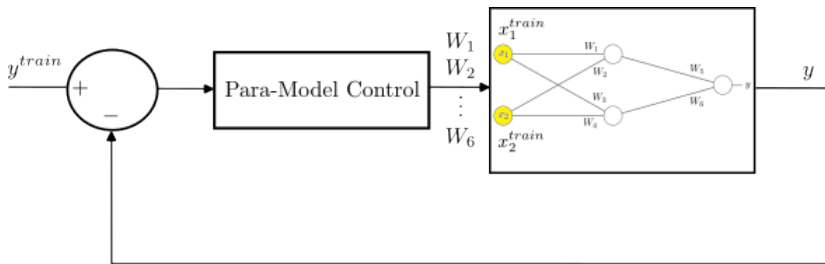


the goal is to adjust the set of the weights W_i to fit the training data using the proposed algorithm such as

$$\text{for all connections } i = 1..6, \quad W_i = PMC(y, y^{train})$$

3. L. Michel, arXiv :1905.02230, 2019 (accepted to APMS'21).

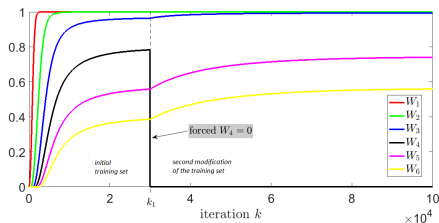
Application to a basic neural network training



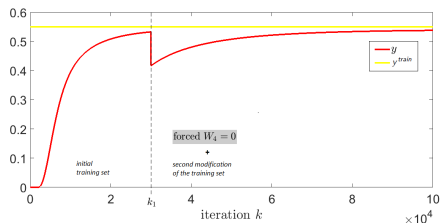
- Enables a priori online updates of the network with respect to topological network modifications & training data changes

Application to a basic neural network training

- Online tuning w.r.t. changes of the training data and the network topology



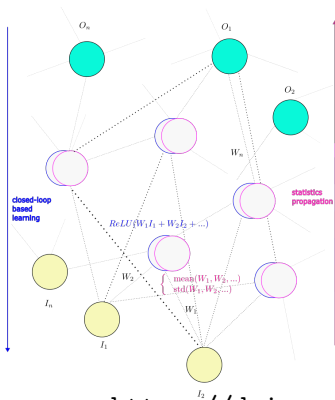
Evolution of the weights W_i



Evolution of the output y

$\Rightarrow$ Allows *a priori* (re)stabilization of the weights W_i according to the different changes of data training and the network topology

Towards growing NN: a connectome perspective^{4 5}



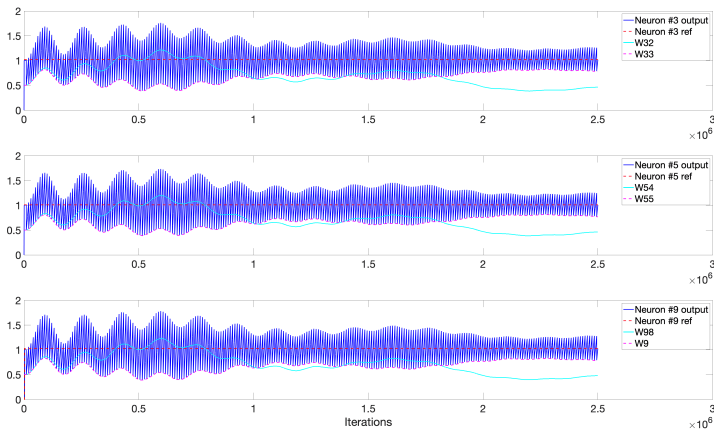
Proposed growing neural network for n inputs (I_n) and n outputs (O_n): to update the synaptic weights $W_1, W_2, \dots, W_n$, the learning process is based on a closed-loop control algorithm and the trained neurons propagate simultaneously some statistical information about the synaptic weights

<https://doi.org/10.5281/zenodo.15072598>

4. J.E. Gaudio et al., IEEE Conference on Decision and Control (CDC), 2019.
5. S. Cerf, E. Rutten, IFAC-PapersOnLine, 2023.

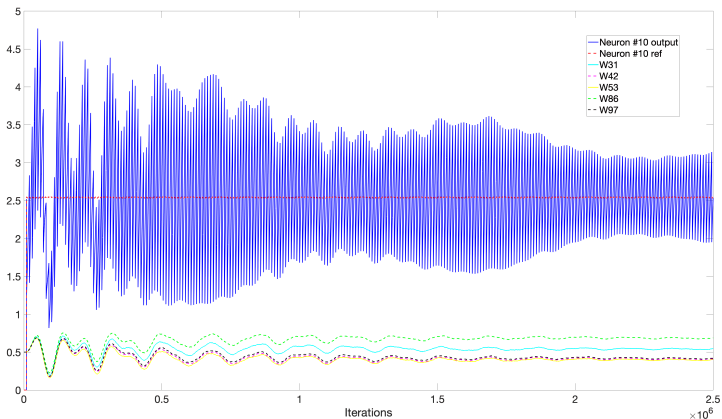
Towards growing NN: a connectome perspective

Consider training 6 neurons (2 inputs and 1 output) to learn, for example, the time-derivative of a sine function



Update of the weights associated to the first neurons #3, #5, #9

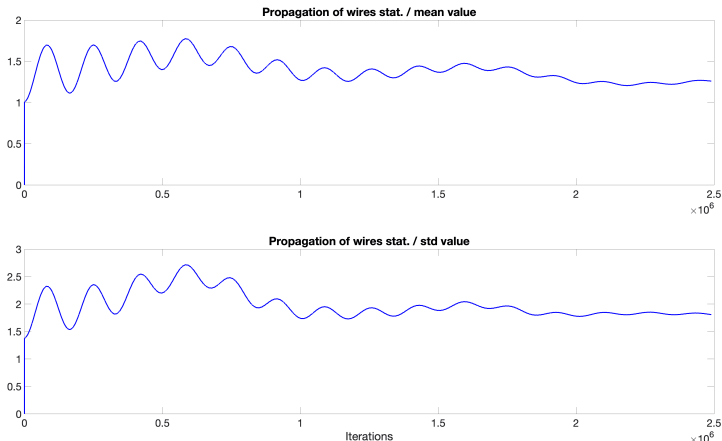
Towards growing NN: a connectome perspective



Update of the weights associated to the output neuron #10

⇒ *Allows a priori stabilization of the weights through the training phase*

Towards growing NN: a connectome perspective



The mean and the standard deviation of the synaptic weights are also propagated, providing additional information to take decisions...

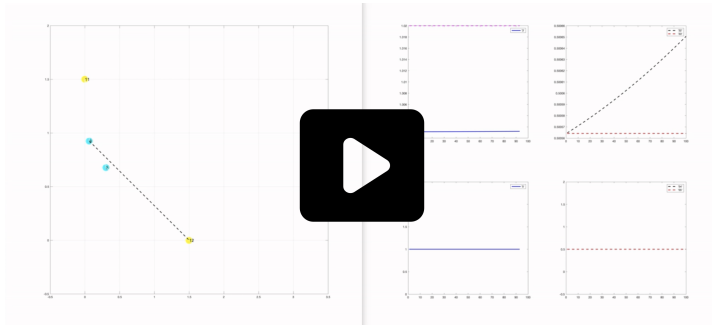
Open perspectives

Very prospective work to be generalized towards large scale neural networks including:

- Cluster analysis of the neural-based information: partition of the learning according to several outputs
- Probabilistic-based spatial interactions between neurons (including e.g. Bayesian inference)
- Introduction to « in-neuron granularity » via connection dynamics
- Improve the neural model (spiking neurons...)
- Emphasize statistics propagation and build framework to help decision-making
- Robustness to large sets of training data

... This work is in progress

Towards growing NN: a connectome perspective



Video showing the neural network growing...

References

- [1] L. Michel, A para-model agent for dynamical systems, arXiv:1202.4707, 2012.

- [2] K. Taras, N. Bredeche and R. Doursat (Editors), Growing Adaptive Machines: Combining Development and Learning in Artificial Neural Networks, Studies in Computational Intelligence, Springer Berlin, Heidelberg, 2014.

- [3] J.E. Gaudio, T.E. Gibson, A.M. Annaswamy, M.A. Bolender and E. Lavretsky, Connections Between Adaptive Control and Optimization in Machine Learning, 2019 IEEE 58th Conference on Decision and Control (CDC), Nice, France, pp. 4563-4568, 2019.

- [4] P. Tabuada, B. Gharesifard, Universal Approximation Power of Deep Residual Neural Networks via Nonlinear Control Theory, arXiv:2007.06007, 2020.

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[5] S. Cerf, E. Rutten, Combining neural networks and control: potentialities, patterns and perspectives, IFAC-PapersOnLine, Volume 56, Issue 2, pp. 9036-9049, 2023.

[6] J. G. Elkins, F. Fahimi, Control-Theoretic Techniques for Online Adaptation of Deep Neural Networks in Dynamical Systems, arXiv:2402.00761, 2024.

[7] K. Itoh, Rethinking Deep Learning: Propagating Information in Neural Networks without Backpropagation and Statistical Optimization, arXiv:2409.03760, 2024.

Thank you for your attention !

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`<https://loicmichelcontrol.github.io/website/>`